

Preface

This book introduces you to the fascinating intersection of biology and deep learning. It's written for both biologists eager to acquire computational skills and computational practitioners curious about applying their expertise to biological problems. This fusion of disciplines is already transforming biotechnology and medicine—and is poised to become foundational across the life sciences.

The material in this book is pitched to be introductory, guiding you from the basics to more intermediate concepts. We aim to balance practical code examples with clear explanations, making new terms and ideas accessible. Real-world Python code appears early and often, helping you develop hands-on intuition. While deep learning is a powerful tool, it's not a one-size-fits-all solution—we emphasize the importance of understanding your data and framing your problem thoughtfully before diving into modeling. We encourage you to start simple, build modular and debuggable code, and add complexity only when it serves a clear purpose.

Although this book is designed for beginners, each chapter builds on the last to develop a practical, end-to-end workflow for applying machine learning to biological data. Our goal is to equip you with tools that are robust enough to solve real-world problems and flexible enough to adapt to your own research questions. In the final chapter, for example, we reproduce key results from a recent *Nature Methods* paper that uses deep learning to uncover spatial protein patterns in microscopy images.

This field evolves incredibly rapidly—libraries change, datasets get updated, and model architectures come in and out of fashion. The core ideas you'll learn—how to structure projects, prepare data, build and evaluate models, and connect predictions to biological questions—will remain useful regardless of the framework or trend.

We use the JAX and Flax deep learning ecosystem throughout, helping you build efficient, flexible models using tools that are increasingly popu-

lar in machine learning research.

Who This Book Is For

This book is for anyone interested in using deep learning to solve biological problems—whether you’re a biologist, a data scientist, or a software engineer. We have aimed to provide an accessible bridge between disciplines and a practical foundation for working in this emerging field. We’ve designed the book to be user-friendly and interdisciplinary, with the goal of supporting readers from a wide variety of backgrounds.

Why We Wrote This Book

The short answer is that there just aren’t many educational resources for people who want to work at the intersection of biology and machine learning. Most materials either focus heavily on one field or assume deep prior knowledge in both. We set out to write the book we wished we had when we were getting started—something approachable, hands-on, and grounded in real biological problems.

While this book is far from perfect (please do send us your suggestions and extensions), we hope it offers a helpful and motivating starting point—and maybe even inspires some readers to pursue a career in this exciting, fast-moving field.

Prerequisites

We assume a basic familiarity with Python and the core scientific Python libraries—namely, NumPy, pandas, and Matplotlib. You don’t need to be an expert programmer, but you should feel comfortable reading and writing basic Python code. If you’re new to Python, there are many excellent tutorials and courses available online to help you get started. We particularly recommend the [Codecademy Python online course](#), and the free book *Automate the Boring Stuff with Python* by Al Sweigart.

Some background in machine learning is also helpful. Concepts like optimization, loss functions, training steps, overfitting, and feedforward neu-

ral networks will come up frequently. If these are unfamiliar, don't worry—brief explanations are included throughout the book, and you can always pause to brush up using external resources as needed.

Why We Focus on Molecular Biology

This book focuses primarily on the “small” side of biology: molecules, cells, and tissues. There are two main reasons for this. First, our backgrounds are in molecular and cellular biology, which naturally influenced the topics we chose. Second, most deep learning applications in biology so far have concentrated on this space, largely due to the availability of large-scale datasets and the quantitative nature of the field.

That said, the techniques and workflows in this book are broadly applicable, and we hope that readers interested in other biological areas—such as ecology, neuroscience, or zoology—will also find inspiration and useful tools here. We're excited to see deep learning methods continue to spread across the life sciences.

Why All the Hands-on Programming Projects

When it comes to machine learning, we believe programming is one of the most effective ways to truly understand what's going on. Theoretical explanations alone can often feel abstract or confusing—especially when complex notation and dense jargon get involved.

Hands-on projects cut through that by making ideas concrete. They offer several distinct advantages:

Active learning

Writing code helps reinforce concepts through direct experience.

Deeper understanding

Implementing a model forces you to understand each component—there's no room for hand-waving.

Immediate feedback

It can be unforgiving, but code that doesn't work quickly highlights gaps in understanding or setup and invites experimentation.

Practical payoff

By the end of a project, you'll have a functioning model you can analyze, adapt, or apply to new data.

Theoretical understanding is still essential—but for most people, there's no substitute for the clarity that comes from building and debugging something yourself.

Notebook Availability

Executable notebooks for each chapter are available on GitHub and can be launched directly in Google Colab. These notebooks are designed to help you follow along with the material in a hands-on way.

We'll remind you where to find them as you go, but we strongly encourage you to dive in—there's no better way to learn than by experimenting and getting your hands dirty. If you spot bugs or have suggestions as you explore the code, feel free to submit a pull request or open an issue on GitHub.

O'Reilly Online Learning Platform

This book is also available through the O'Reilly online learning platform, where you'll find all figures in full color. The print version is in black and white, and while we've done our best to ensure clarity, some visualizations do rely on color to be fully interpretable.

We recommend referring to the digital version when viewing figures that use heatmaps, overlays, or color-coded labels.

Quick Tour of the Book

This is a highly interdisciplinary book built around end-to-end examples of deep learning applications in biology. Each chapter walks you through a self-contained project—from biological background to dataset construction, model training, evaluation, and possible extensions. While some chapters build on earlier ones, most can be read independently based on your interests.

To support learning, each chapter includes a brief biology and machine learning primer relevant to the task at hand. We cover a wide range of biological modalities (e.g., sequences, images, and graphs) and deep learning architectures (e.g., transformers, convolutional neural networks, graph neural networks, autoencoders), gradually increasing in complexity.

Here's a summary of the five main technical chapters to give you a quick sense of the tasks, data types, and techniques explored:

Chapter	Title	Task	Modality	Task Type	Architecture	Difficulty
2	“Learning the Language of Proteins”	Predict protein function from protein sequence.	Sequence	Multilabel classification	Linear classifier on transformer embeddings	Beginner
3	“Learning the Logic of DNA”	Predict DNA–protein binding events from sequence.	Sequence	Binary and multilabel classification	Convolutional neural network, transformer	Intermediate
4	“Understanding Drug–Drug Interactions Using Graphs”	Predict whether a pair of drugs will interact.	Graph	Graph-based learning, binary classification	Graph neural network	Intermediate
5	“Detecting Skin Cancer in Medical Images”	Classify skin lesions from images.	Image	Multiclass classification	Convolutional neural network	Intermediate
6	“Learning Spatial Organization Patterns Within Cells”	Predict protein subcellular localization.	Image	Representation learning, multilabel classification	Autoencoder, convolutional neural network	Advanced

The chapters generally build in scope and depth, starting with a simple linear classifier on extracted embeddings in [Chapter 2](#) and progressing to more complex models built from scratch on increasingly rich and challenging data.

Chapter Summaries

[Chapter 1, “Introduction”](#)

Learn about the promise and challenges of deep learning in biology. We walk through practical questions to consider before launching a new project—like what your model could replace, whether deep learning is even necessary, and how to structure your workflow.

This chapter also includes a short technical introduction covering JAX/Flax, Python patterns common in machine learning, working environments, and practical setup tips.

Chapter 2, “Learning the Language of Proteins”

Use protein language model embeddings to predict protein function. You’ll extract embeddings from a pretrained protein language model and train a simple linear classifier to perform multilabel classification. Along the way, you’ll learn how to prepare protein function datasets, evaluate predictions, and visualize embeddings to gain biological insight.

Chapter 3, “Learning the Logic of DNA”

Train lightweight convolutional and transformer models to predict where certain regulatory proteins bind to DNA. You’ll learn about how to uncover insights from DNA sequences, interpret models, and evaluate predictions across multiple binding factors.

Chapter 4, “Understanding Drug–Drug Interactions Using Graphs”

Use graph neural networks to model drug relationships, predicting potential drug interactions—harmful or beneficial—based on network features.

Chapter 5, “Detecting Skin Cancer in Medical Images”

Train convolutional neural networks to classify types of skin lesions from images. This chapter focuses on image preprocessing, transfer learning, and managing overfitting in small medical datasets.

Chapter 6, “Learning Spatial Organization Patterns Within Cells”

Build a variational autoencoder to analyze protein localization from microscopy images. This advanced chapter explores unsuper-

vised learning, latent space analysis, and extracting structure from high-resolution imaging data.

Chapter 7, “Tips and Tricks for Deep Learning in Biology”

Gain practical advice for designing, debugging, and improving models in real biological workflows. This chapter also provides guidance on avoiding common pitfalls, extending your learning, and exploring new project ideas.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

Constant width

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

Using Code Examples

Supplemental material (code examples, exercises, etc.) is available for download at <https://github.com/deep-learning-for-biology>.

If you have a technical question or a problem using the code examples, please open an issue, submit a pull request on GitHub, or email support@oreilly.com.

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We have a web page for this book, where we list errata, examples, and any additional information. You can access this page at <https://oreil.ly/deep-learning-for-biology-1e>.

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Acknowledgments

This book was a monumental project—one we could not have completed in isolation—and we are deeply grateful to everyone who supported us

along the way. We especially thank Melis Kayıkcı and Tibet Fonteyne and our families for their unwavering support. This book would not have been possible without them.

We are also grateful to our technical reviewers—Christian Winkler and Lipi Deepaakshi Patnaik—whose thoughtful feedback helped sharpen both the clarity and correctness of the material. We would also like to thank Danny Elfanbaum for developing the `jupyter-book-to-htmlbook` tool, which greatly improved the content development. Another big thank-you to our O'Reilly editor, Corbin Collins, for guiding the project from start to finish; production editor Ashley Stussy, editors Audrey Doyle and Kim Wimpsett, and WordCo Indexing Services, Inc for helping to polish the manuscript; and acquisitions editor Nicole Butterfield, for encouraging us to take on this project in the first place.

Many friends and colleagues offered insightful reviews, advice, and much-needed encouragement throughout the writing process. Our sincere thanks go to Sam Beckbessinger, Petar Veličković, Justin Dollman, Kristofer Linton-Reid, Felix Kruger, Joshua Pan, Richard Tanburn, Sebastian Bodenstein, Arnaud Aillaud, Vaibhav Bhardwaj, Fredrik Eckardt, Hannes Baukmann, Justin Cope, Levent Mengütürk, Murat Mengütürk, James Sullivan, Loïc Royer, Toby Pohlen, Žiga Avsec, and John Jumper.

Finally, Chapter 2 was inspired by the 2023 Deep Learning Indaba practical “Machine Learning for Biology: Learning the Language of Life,” co-developed by Tom Makkink, Kevin Michael Eloff, Natasha Latysheva, and Kyle Taylor.